# Exercise 4 Union and Union ALL

## Question 1 (Union)

1) Unique Account Holders:

```
-- Unique list of account holders across Sandton and Rosebank
SELECT account_id, account_holder, 'Sandton' As city
FROM branch_sandton_accounts
UNION
SELECT account_id, account_holder, 'Rosebank' As city
FROM branch_rosebank_accounts;
```

Output table:

| account_id | account_holder | city |
|---|---|---|
| A001 | Nomvula Dlamini | Sandton |
| A002 | David Mokoena | Sandton |
| A003 | Lerato Sithole | Sandton |
| A004 | Peter Nkosi | Sandton |
| A005 | Zanele Khumalo | Rosebank |
| A006 | Thabo Motha | Rosebank |

2) Product Catalogue:

```
-- Unique product catalogue combining savings and current products
SELECT product_code, product_name, 'savings' As product_type
FROM savings_products
UNION
SELECT product_code, product_name, 'current' As product_type
FROM current_products;
```

Output Table:

| Product_code | product_name | product_type |
|---|---|---|
| SV01 | Basic savings | Savings |
| SV02 | Premium Savings | Savings |
| SV03 | Youth Savings | Youth Savings |
| SV04 | Buriness savings | Savings |
| CR01 | Standard current | Current |
| CR02 | Gold Current | Current |
| CR03 | Buriness Current | Current |

3) Internal Bank Staff

```
-- Unique list of bank employees across divisions
SELECT staff_id, staff_name, LOWER(REPLACE(staff_name, ' ', '.')) || '@nexbank.co.za' AS email
FROM retail_banking_staff
UNION
SELECT staff_id, staff_name, LOWER(REPLACE(staff_name, ' ', '.')) || '@nexbank.co.za' AS email
FROM corporate_banking_staff;
```

| staff_id | staff_name | email |
|---|---|---|
| S001 | Mpho Radebe | mpho.radebe@nexbank.co.za |
| S002 | Brian Tshabalala | brian.tshabalala@nexbank.co.za |
| S003 | Aisha Patel | aisha.patel@nexbank.co.za |
| S004 | Kabelo Mogabelo | kabelo.mogabelo@nexbank.co.za |
| S005 | Nandi Dube | nandi.dube@nexbank.co.za |
| S006 | Sipho Khumalo | sipho.khumalo@nexbank.co.za |

## 4) Branch Footprint

-- Unique list of cities where Nexbank serves customers

```
SELECT city_code, city_name, 'Mobile' As region
FROM mobile_branch_cities
UNION
SELECT city_code, city_name, 'Digital' As region
FROM digital_branch_cities;
```

| city_code | city_name | region |
|---|---|---|
| C01 | Johannesburg | Gauteng |
| C02 | Pretoria | Gauteng |
| C03 | Cape Town | Western Cape |
| C04 | Durban | KwaZulu-Natal |
| C05 | Polokwane | Limpopo |
| C06 | Port Elizabeth | Eastern Cape |

## 5) Campaign Targets

-- Unique list of customers targeted in recent marketing campaigns

```
SELECT customer_id, customer_name, 'Push Notifi-
cation' As segment
FROM push_notification_targets
UNION
SELECT customer_id, customer_name, 'In-App Banner'
As segment
FROM inapp_banner_targets;
```

| Customer_id | customer_name | segment |
|---|---|---|
| C1001 | Nomsa Zwane | Core |
| C1002 | Andile Buthelezi | Mass |
| C1003 | Fatima Mahomed | Premium |
| C1004 | Ryno van Zyl | Youth |
| C1005 | Thandeka Cele | Mass |
| C1006 | Samuel Nkosi | Core |

# Part 2: (UNION ALL)

## 6) ATM Logs Reconciliation

```
-- Full logs combined from ATM-01 and ATM-02 machines
SELECT txn_id As transaction_id, account_id, amount,
CURRENT_DATE As transaction_date
FROM atm01_transactions
UNION ALL
SELECT txn_id As transaction_id, account_id, amount,
CURRENT_DATE As transaction_date
FROM atm02_transactions;
```

| transaction_id | account_id | amount | transaction_date |
|---|---|---|---|
| T1001 | A001 | 500.00 | 2026-10-23 |
| T1002 | A002 | 1200.00 | 2026-10-23 |
| T1003 | A001 | 300.00 | 2026-10-23 |
| T1004 | A003 | 750.00 | 2026-10-23 |
| T1003 | A001 | 300.00 | 2026-10-23 |
| T1005 | A004 | 900.00 | 2026-10-23 |
| T1006 | A002 | 450.00 | 2026-10-23 |
| T1007 | A005 | 150.00 | 2026-10-23 |

## 7) Credit Risk Evaluation

```
-- Complete list of loan applications from Gauteng and Western Cape
SELECT app_id As application_id, customer_id, loan_type,
NULL As amount_requested
FROM gauteng_loan_applications
UNION ALL
SELECT app_id As application_id, customer_id, loan_type,
NULL As amount_requested
FROM Western_cape_loan_applications;
```

| application_id | customer_id | loan_type | amount_requested |
|---|---|---|---|
| LA001 | C1001 | Home Loan | NULL |
| LA002 | C1002 | Vehicle Loan | NULL |
| LA003 | C1003 | Personal | NULL |
| LA004 | C1004 | Home Loan | NULL |
| LA003 | C1003 | Personal | NULL |
| LA005 | C1005 | Vehicle Loan | NULL |
| LA006 | C1006 | Business | NULL |
| LA007 | C1007 | Home Loan | NULL |

## 8) Escalations Log

-- combined feed of email and app complaints for total volume analysis

```
SELECT complaint_id, customer_id, 'Email' As category,
CURRENT_DATE AS logged_date
FROM email_complaints
UNION ALL
SELECT complaint_id, customer_id, 'App' As category,
CURRENT_DATE AS logged_date
FROM app_complaints;
```

| complaint_id | customer_id | category | logged_date |
|---|---|---|---|
| EC001 | C2001 | Email | 2026-10-23 |
| EC002 | C2002 | Email | 2026-10-23 |
| EC003 | C2003 | Email | 2026-10-23 |
| EC004 | C2004 | Email | 2026-10-23 |
| AC001 | C2005 | App | 2026-10-23 |
| AC002 | C2001 | App | 2026-10-23 |
| AC003 | C2006 | App | 2026-10-23 |
| AC004 | C2007 | App | 2026-10-23 |

9) Financial Statements

-- consolidated multi-month payment records for reconciliation

```
SELECT payment_id, account_id, amount, 'April' AS
payment_date
FROM april_payments
UNION ALL
SELECT payment_id, account_id, amount, 'May' AS
payment_date
FROM may_payments;
```

| payment_id | account_id | amount | payment_date |
|---|---|---|---|
| PAY001 | A001 | 12500.00 | April |
| PAY002 | A002 | 4800.00 | April |
| PAY003 | A003 | 9200.00 | April |
| PAY004 | A004 | 3300.00 | April |
| PAY005 | A001 | 12500.00 | May |
| PAY006 | A005 | 7600.00 | May |
| PAY007 | A002 | 5100.00 | May |
| PAY008 | A006 | 2800.00 | May |

10) General Ledger Balances

-- combined debit and credit lines for full financial ledger transparency

```
SELECT entry_id, account_id, entry_type, NULL AS amount,
CURRENT_DATE AS entry_date
FROM debit_entries
UNION ALL
SELECT entry_id, account_id, entry_type, NULL AS amount,
CURRENT_DATE AS entry_date
FROM credit_entries;
```

| entry_id | account_id | entry_type | amount | entry_date |
|---|---|---|---|---|
| DR001 | A001 | Debit | NULL | 2026-10-23 |
| DR002 | A002 | Debit | NULL | 2026-10-23 |
| DR003 | A003 | Debit | NULL | 2026-10-23 |
| DR004 | A004 | Debit | NULL | 2026-10-23 |
| CR001 | A001 | Credit | NULL | 2026-10-23 |
| CR002 | A005 | Credit | NULL | 2026-10-23 |
| CR003 | A002 | Credit | NULL | 2026-10-23 |
| CR004 | A006 | Credit | NULL | 2026-10-23 |

## PART 3 Bonus Questions

1) The analyst should use the UNION operator because the objective here is to generate a distinct master list of individuals spanning both months. If a client is high-risk in both January and February, standard UNION ensures that they only appear once on the master list. By choosing standard UNION, the database automatically deduplicates the results, ensuring that every high-risk customer shows up once.

2) The auditing team should use UNION ALL, because for auditing and tracking absolute transaction volume, every single entry is critical. If two identical transactions occur across different sub-systems, standard UNION would treat the second one as a duplicate and delete it from the results. This would artificially lower the transaction metrics. UNION ALL ensures no data is dropped and everything is ~~accounted~~ for. accounted for.

3) The query crashes because the two SELECT statements don't have the same number of columns. SQL requires structural symmetry to merge datasets; the Sandton query only pulls two fields (account_id, account_holder), while the Rosebank query pulls three (account_id, account_holder, city).

-How to fix?: This can be resolved by adding a null placeholder to the first query so both sides match up perfectly.

Sql:

```
SELECT account_id, account_holder, NULL AS city
FROM branch_sandton_accounts
UNION
SELECT account_id, account_holder, city FROM
branch_rosebank_accounts;
```